

Does the play-in tournament help the best team win the title?

A model-based counterfactual over five NBA seasons, 2020-21 to 2024-25

Generated 21 September 2026 from `NBA_PlayIn_Whiteboard_Data.xlsx`. Every figure in this report is produced by `python run_all.py`; nothing is typed in by hand.

1. Summary

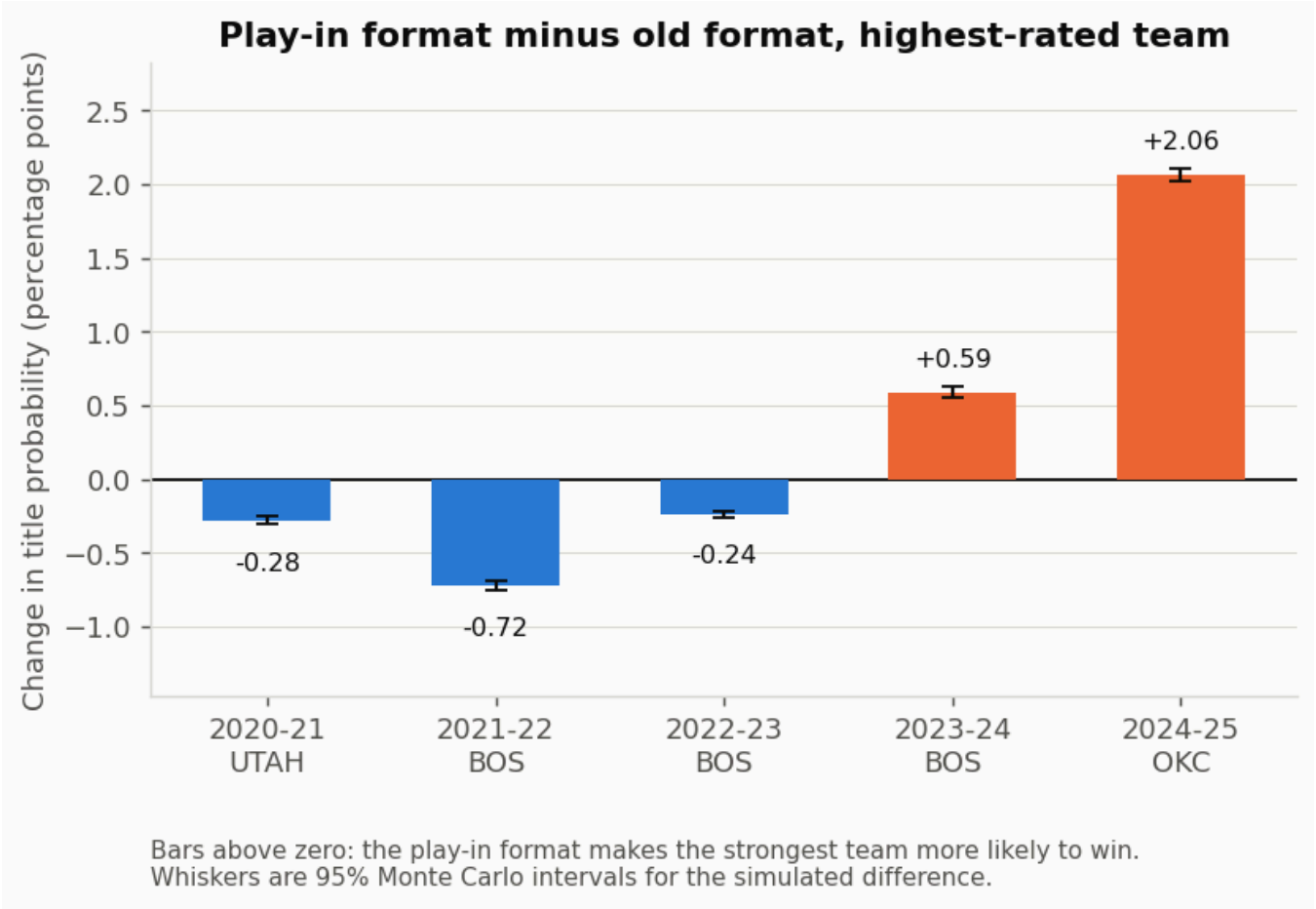
Holding each season's standings and team ratings fixed, the play-in tournament changed the highest-rated team's probability of winning the championship by **+0.29 percentage points on average over the five seasons** (simulated: +0.28 pp, Monte Carlo standard error 0.008 pp). The effect was positive in 2 of the five seasons and negative in 3. Every season's effect is smaller than one percentage point except 2024-25, at +2.08 pp.

The sign is not a property of the play-in format. It depends on one thing: whether the play-in tends to put a **stronger or weaker** team into the seed that the best team has to beat.

- In **2024-25** the strongest team, OKC, finished first in the West and so faced the eighth seed. Under the old format that was Memphis, rated 4.79, the strongest of the four play-in teams. Under the play-in format the eighth seed is Memphis only 35.5% of the time, and its expected rating falls to 2.83. OKC's draw gets easier, and its title probability rises +2.08 pp.
- In **2021-22** the strongest team, Boston, finished second in the East and so faced the seventh seed. Under the old format that was Brooklyn, the weaker of the two; the play-in replaces it with the winner of Brooklyn against a better-rated Cleveland. Boston's draw gets harder, and its title probability falls 0.71 pp.

Two other model-derived quantities are reported alongside: the expected rating of the champion and the expected rating of the sixteen-team playoff field. Both move by small amounts, and like the headline number they move in different directions in different seasons.

This is a counterfactual inside a model, not the causal effect of the policy. Standings, ratings and the game model are held fixed, so nothing about incentives, resting, injuries or roster decisions is represented. See section 12.



2. Research question and scope

Holding regular-season standings and estimated team strengths fixed, does the NBA play-in tournament increase the probability that the highest-rated team wins the championship?

Five seasons are studied: 2020-21, 2021-22, 2022-23, 2023-24, 2024-25. For each one, the same standings, the same ratings and the same game-probability model are put through two tournaments:

	Old qualification format	Play-in format
Seeds 1-6	positions 1-6 qualify	positions 1-6 qualify
Seed 7	position 7	winner of 7 hosts 8
Seed 8	position 8	winner of (7/8 loser hosts 9/10 winner)
Teams involved	8 per conference	10 per conference

"Strongest" always means the highest full-season SRS rating computed **before** that season's postseason. The actual champion is never used to define it.

What the comparison can and cannot show

It isolates one channel: the bracket. Because the standings and ratings are frozen, the comparison answers "given these teams and these strengths, what does the extra round do to the best team's chances?" It does not answer "what happened to the NBA because the play-in exists?", since a real policy also changes how teams behave during the season.

3. Data and audit

Sources

Source	Use	Retrieved
NBA_PlayIn_Whiteboard_Data.xlsx	the study's input: 9,519 regular-season games 2017-18 to 2024-25, 150 team-seasons, published reference values	supplied
sportsdataverse/hoopR-nba-data schedule archive	independent rebuild of the game table	2026-09-21
ESPN standings API	win-loss records and scoring totals, five seasons	2026-09-21
ESPN game summary API	box scores for 21 individual games under investigation	2026-09-21
Basketball-Reference season totals	third source for the 12 disputed scoring totals	2026-09-21
NBA.com play-in rules	format verification	2026-09-21

Audit findings

Check	Status	What was found
Game IDs unique	pass	9519 games, 0 duplicated IDs
Games per season	pass	counts 2018:1230, 2019:1230, 2020:1059, 2021:1080, 2022:1230, 2023:1230, 2024:1230, 2025:1230; all match the published schedule
No repeated date/home/away pairs	pass	0 repeated matchup-days
One team plays at most one game per day	known issue	1 team-day(s) hold two games: game IDs 401161536, 401161543. Game 401161536 (Golden State 115, Phoenix 99) is dated 2020-03-01 in the archive but was played on 2020-02-29 (confirmed against Basketball-Reference box-score index). The score is correct; only the date is off by one, and it affects nothing but the day on which that 2019-20 result enters the running ratings.
Team identities	pass	30 distinct team codes: ATL BKN BOS CHA CHI CLE DAL DEN DET GS HOU IND LAC LAL MEM MIA MIL MIN NO NY OKC ORL PHI PHX POR SA SAC TOR UTAH WSH
30 teams every season	pass	min 30, max 30
Season labels match dates	pass	2018: 2017-10-17..2018-04-11; 2019: 2018-10-16..2019-04-10; 2020: 2019-10-22..2020-08-14; 2021: 2020-12-22..2021-05-16; 2022: 2021-10-19..2022-04-10; 2023: 2022-10-18..2023-04-09; 2024: 2023-10-24..2024-04-14; 2025: 2024-10-22..2025-04-13
Cup championship games excluded, other	pass	both Cup finals absent (2023 In-Season Tournament final (LAL 123, IND 109); 2024 Cup final (MIL 97, OKC 81)); 132 of 132 other Cup games retained, because they do count

Check	Status	What was found
Cup games kept		toward the regular-season standings
No All-Star or postseason games in the regular-season table	pass	0 non-regular-season game IDs found in the sheet
Workbook game table reproduced from the hoopR archive	pass	9519 games matched; 0 only in the workbook, 0 only in the archive, 0 score differences, 0 date differences, 0 neutral-flag differences
2020 bubble games treated as neutral	pass	88 bubble games, 88 neutral in the model; source flag marked 0 of them neutral
Other neutral-site games	info	17 non-bubble neutral games at 7 venues (see audit_neutral_sites.csv)
Win-loss records match ESPN standings	pass	150 team-seasons checked against a freshly retrieved standings feed; 0 mismatches
Scoring totals: game log vs standings feed	known issue	12 of 150 team-seasons differ, by 1-3 points; the workbook reports the same 12
Third source (Basketball-Reference) on the 12 scoring gaps	resolved	12 of 12 team-season totals match the summed game scores exactly, so the ESPN standings aggregate is the outlier and no game score needs correcting
Scoring discrepancies traced to individual games	resolved	21 candidate games checked against ESPN box scores; 0 could not be checked
Regular-season positions 1-15 restored	pass	ranks 1-15 complete in all 10 conference-seasons; 40 positions 7-10 confirmed from opening play-in matchups; 17 team-seasons where the stored ESPN playoff seed differs from the regular-season position
No future information in the pregame features	pass	7048 published feature rows; prior-game counts recomputed from the game table: 0 disagreements, 0 rows below the 20-game threshold

The twelve scoring discrepancies, resolved

Twelve of 150 team-season scoring totals in the ESPN standings feed differ from the sum of that team's game scores, by one to three points. They are not scattered: they pair up. In 2020-21, Indiana's points-against is two higher in the game log while Dallas's points-for is two higher by the same amount, which can only come from a game between those two teams. That logic narrows the whole set of twelve to 21 candidate games.

All 21 candidate box scores were pulled from ESPN's game summary endpoint: **every one agrees with the game table**. Basketball-Reference was then checked as a third, fully independent source: all 12 team-season totals match the summed game scores exactly, not the standings feed.

Conclusion: the game scores are right and the ESPN standings aggregate is the outlier. No score is corrected, no game is dropped, and the analysis uses the game-level scores. Section 11 shows what would happen if the standings totals were used instead: at most 0.020 pp on any season's result.

One archive defect found

Game [401161536](#) (Golden State 115, Phoenix 99) is dated 2020-03-01 in the archive, which puts two Golden State games on the same day. Basketball-Reference's box-score index shows the game was played 2020-02-

29. The score is right; the date is one day late. It sits in 2019-20, a training season only. Moving it to its true date changes no season's result by more than 0.042 thousandths of a percentage point.

Exclusions and venue handling

- Only completed regular-season games between the 30 NBA teams are kept.
- The two in-season Cup **championship** games are removed, because they do not count in the standings. Every other Cup game is kept, because it does.
- No All-Star or postseason game appears in the regular-season table.
- The 88 games played in the 2020 Orlando bubble are treated as neutral. The source feed marks none of them neutral, which is wrong; the override is the workbook's and it is correct.
- Positions 7-10 are recovered from the wording of the opening play-in matchups ("7th Place vs 8th Place"), not from the playoff seeds, which the play-in itself changes. In 17 team-seasons the stored playoff seed differs from the regular-season position, so this matters.

4. Ratings

Full-season SRS

For every team i , with $mbar_i$ its average scoring margin, n_i its number of games and O_i the list of its opponents (one entry per game played):

$$R_i = mbar_i + (1/n_i) * \text{sum over } j \text{ in } O_i \text{ of } R_j, \quad \text{sum over } i \text{ of } R_i = 0$$

Multiplying through by n_i turns this into a linear system, $(\text{diag}(N) - C) R = m$, where $C[i,j]$ counts the games between i and j and m_i is team i 's total scoring margin. It is solved with the sum-to-zero constraint. The rating is a team's scoring margin adjusted for how strong its opponents were. A repeated opponent is counted once per game, which is what makes the adjustment right for unbalanced schedules.

These ratings reproduce the workbook's to within $1e-13$, and they satisfy their own defining equation to within $1e-7$ (tested in `tests/test_ratings.py`).

Pregame ratings

The same system is re-solved once per game date using only games on **strictly earlier** dates, so a game can never help predict itself. A game enters the training set only if both teams already have 20 games that season, which keeps ratings built on five-game samples out of the fit. Of 9,519 games, 7,048 pass that filter. Recomputed independently here, all 7,048 match the workbook's published features exactly.

That threshold is a real choice with a cost: it throws away roughly the first six weeks of every season. It is kept because early ratings are extremely noisy - before 20 games a team's SRS still moves by several points per game - and because it was specified before any result was looked at.

5. The game model

$$P(A \text{ beats } B) = 1 / (1 + \exp(-(\text{beta} * (R_A - R_B) + h * H)))$$

$H = +1$ if A is at home, -1 if A is away, 0 at a neutral venue

Fitted by maximum likelihood on completed games from earlier seasons only, with $\text{beta} \geq 0$ and no separate intercept, so two equally rated teams at a neutral venue are a coin flip by construction. The prespecified training window is the **three completed seasons before** the season being studied; one- and two-season windows are reported in section 11 as checks, never as a choice.

Season	Highest-rated team	SRS	Position	Conference	beta	h	Training games
2020-21	UTAH	8.96	1	Western	0.1148	0.3378	2,593
2021-22	BOS	7.02	2	Eastern	0.1102	0.2918	2,441
2022-23	BOS	6.38	2	Eastern	0.1077	0.2000	2,441
2023-24	BOS	10.74	1	Eastern	0.1114	0.2439	2,613
2024-25	OKC	12.70	1	Western	0.1129	0.2225	2,767

beta is close to 0.111 in every season: one point of SRS is worth about 0.111 in log-odds, so a five-point rating edge makes a team roughly 64% to win a neutral game. h between 0.20 and 0.34 means home court is worth between 1.9 and 2.9 points of rating - in the same range as the familiar "home court is worth about two or three points" rule.

No model was chosen because it produced a preferred tournament answer. The window was fixed in advance, and the alternatives are reported whatever they show.

6. The two tournaments

Play-in, exactly as the league runs it

Position 7 hosts position 8; the winner is **seed 7**. Position 9 hosts position 10; the loser is eliminated. The loser of 7/8 hosts the winner of 9/10; that winner is **seed 8**. (Verified against nba.com.)

This produces six possible ordered (seed 7, seed 8) outcomes per conference, and they are **dependent**: whoever wins the 7/8 game is seed 7 and cannot also be seed 8. The study carries the joint distribution over all six, never two independent marginals. `tests/test_structure.py` contains a test that fails if anyone replaces the joint with a product.

Bracket and venues

The bracket is fixed and there is no reseeding: 1-8, 4-5, 2-7, 3-6; the winners of 1/8 and 4/5 meet, as do the winners of 2/7 and 3/6. The better seed hosts game 1 of every conference series, in the 2-2-1-1-1 pattern H, H, A, A, H, A, H .

Finals home court

The published NBA rule for two teams in opposite conferences is applied in order: better regular-season winning percentage; then head-to-head record; then record against the opposite conference. Across the five seasons there are 24 tied pairs among teams that could reach the Finals, and the chain resolves **all of them** - 12 on head-to-head and 12 on opposite-conference record. None is left unresolved, so the workbook's 50/50 pilot assumption is not needed anywhere.

Section 11 shows that this matters even less than it sounds: the highest-rated team is never itself in a tied pair, so its title probability is **bit-for-bit identical** under all four hosting rules tried.

7. Exact calculation

Nothing in this section is simulated.

Series. With p_k the probability the tracked team wins game k , and $F(w, l)$ the probability it wins the series from w wins and l losses:

$$\begin{aligned} F(w, l) &= p_{w+l} * F(w+1, l) + (1 - p_{w+l}) * F(w, l+1) \\ F(4, l) &= 1, \quad F(w, 4) = 0 \end{aligned}$$

This is exact for venue-dependent probabilities, which a binomial formula is not: the games in a series are not all played under the same conditions.

Bracket. Each round's output is a probability distribution over which team occupies the slot. Two slot distributions are combined by summing, over all pairs, the probability that they meet times the probability that each wins. After three rounds the conference winner's distribution remains; the two conferences are then crossed, pair by pair, with the Finals hosting rule.

Play-in. Six ordered seed states per conference are enumerated, and the 36 combined configurations are crossed. The title distribution is the weighted average.

Every distribution is checked to sum to one, and the four play-in teams' qualification probabilities are checked to sum to exactly two per conference.

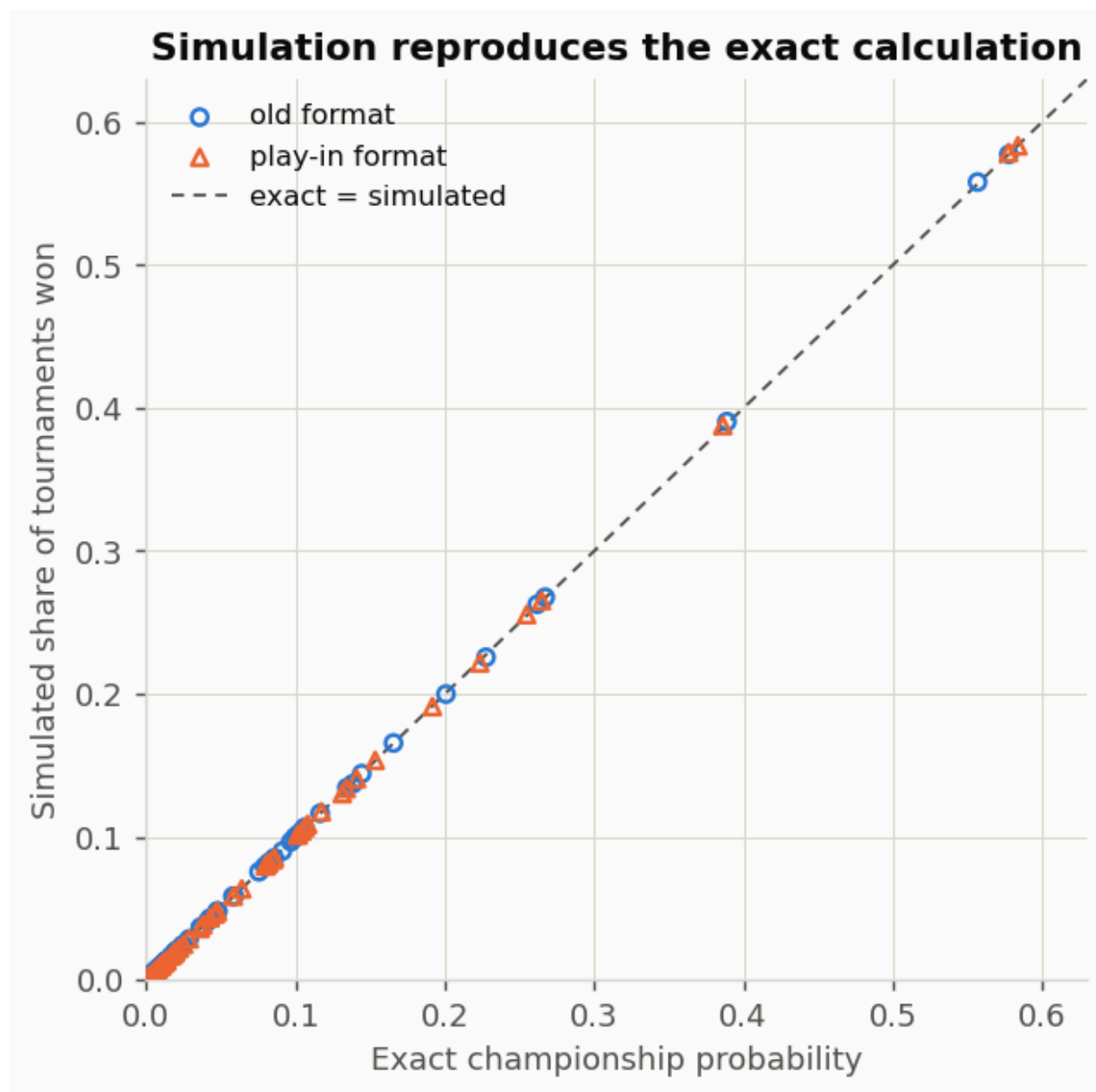
8. Monte Carlo design

The simulator plays games, not tournaments drawn from the answer. It draws uniforms, compares them with game probabilities, counts series wins, and carries the winners forward through exactly the same bracket rules - so agreement with section 7 is evidence that both implementations are right.

- **500,000 complete tournaments per format per season** (5,000,000 in total), configurable with `--replicates`.
- Seed 20250921. Replicates are drawn in fixed blocks of 20,000, each block seeded from (seed, season, block number), so the same seed reproduces the same run on any machine and asking for more replicates extends the run instead of changing it.
- **Common random numbers.** Both formats use the same draws for the same bracket position, so the two runs are paired. The reported standard error for the difference is the standard error of the paired differences, `sd(d) / sqrt(M)`, which is the correct one for this design. The independent-proportions standard error is also reported: it is about 6 times larger, which is the value of pairing.
- A best-of-seven is simulated by playing all seven games and awarding the series to whoever wins at least four. That is exactly equivalent to stopping at four wins, because the winner of the first four games is always the winner of all seven.

Simulation against exact calculation

Across all 30 teams, five seasons and both formats (300 comparisons), the largest gap between simulated share and exact probability is 0.00186 and the mean gap is 0.000132. The largest gap in units of its own Monte Carlo standard error is 3.13, which is what 300 independent comparisons should produce. No seed was changed to make these agree; the seed was fixed before the first run.



Monte Carlo error is computational uncertainty only. It says how precisely the simulation reproduces the model's own answer. It says nothing about whether the ratings are right, whether the logistic model is the right model, or what the true effect of the policy is.

9. Validation

Held out means held out: for a season's test set, that season's games never enter the fit.

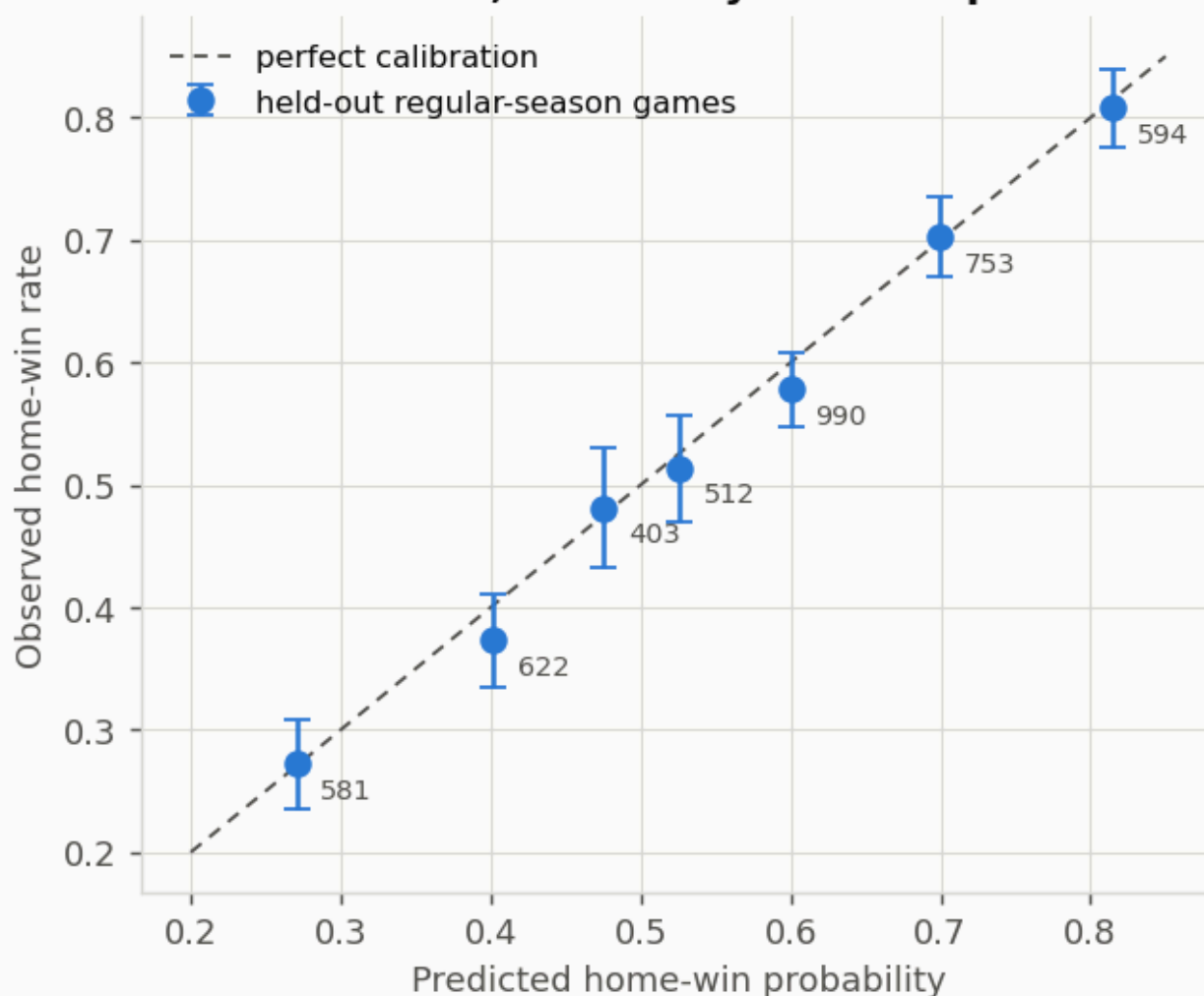
Test set	Games	Brier, model	Brier, home-rate baseline	Log loss, model	Log loss, baseline	Accuracy, model	Accuracy, baseline
Regular season (pregame ratings)	4455	0.2189	0.2482	0.6268	0.6896	0.643	0.545
Play-in games (frozen ratings)	30	0.2218	0.2323	0.6352	0.6576	0.667	0.667
Playoffs (frozen ratings)	422	0.2304	0.2441	0.6530	0.6813	0.637	0.583

The model beats the naive home-win-rate baseline on Brier score and log loss on all three test sets, and by a wider margin in the regular season, where rating differences are largest. Its accuracy advantage over the baseline is about 10 percentage points in the regular season and 5 in the playoffs. The playoff margin is thinner, which is expected: playoff teams are closer in strength than a random pair of NBA teams.

The baseline is a real baseline, not a straw man - it is fitted on the same training window and predicts the historical home-win rate (0.555 on average) for every hosted game.

Calibration

Calibration, five study seasons pooled



Predicted from	to	Games	Mean predicted	Observed home-win rate	Standard error
0.0000	0.3500	581	0.2704	0.2719	0.0185
0.3500	0.4500	622	0.4020	0.3730	0.0194
0.4500	0.5000	403	0.4753	0.4814	0.0249
0.5000	0.5500	512	0.5255	0.5137	0.0221
0.5500	0.6500	990	0.5999	0.5778	0.0157
0.6500	0.7500	753	0.6996	0.7025	0.0167
0.7500	1.0000	594	0.8149	0.8081	0.0162

Predicted and observed home-win rates track each other across the whole range; the biggest single-bin gap is 0.029, which is 1.5 standard errors. A model used to multiply many probabilities together through a bracket has to be calibrated, not merely accurate, and this one is.

10. Results

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Season	Highest-rated team	Old-format simulated title probability	Play-in simulated title probability	Difference in percentage points
2020-21	UTAH	39.05%	38.77%	-0.28
2021-22	BOS	26.27%	25.55%	-0.72
2022-23	BOS	26.76%	26.52%	-0.24
2023-24	BOS	57.74%	58.33%	+0.59
2024-25	OKC	55.80%	57.87%	+2.06

Five-season average effect, equal weight per season: **+0.28 percentage points** (exact calculation: +0.29 pp).

Counts, exact probabilities and Monte Carlo intervals

Season	Team	M (tournaments per format)	W old	W play-in	Exact old	Exact play-in	Exact difference (pp)	Simulated difference (pp)	95% interval (pp)
2020-21	UTAH	500,000	195,232	193,854	0.38888	0.38612	-0.276	-0.276	[-0.306, -0.245]
2021-22	BOS	500,000	131,365	127,751	0.26188	0.25478	-0.710	-0.723	[-0.754, -0.692]
2022-23	BOS	500,000	133,805	132,621	0.26716	0.26479	-0.237	-0.237	[-0.259, -0.214]
2023-24	BOS	500,000	288,707	291,655	0.57772	0.58378	+0.605	+0.590	[+0.550, +0.629]
2024-25	OKC	500,000	279,012	289,328	0.55678	0.57756	+2.079	+2.063	[+2.020, +2.106]

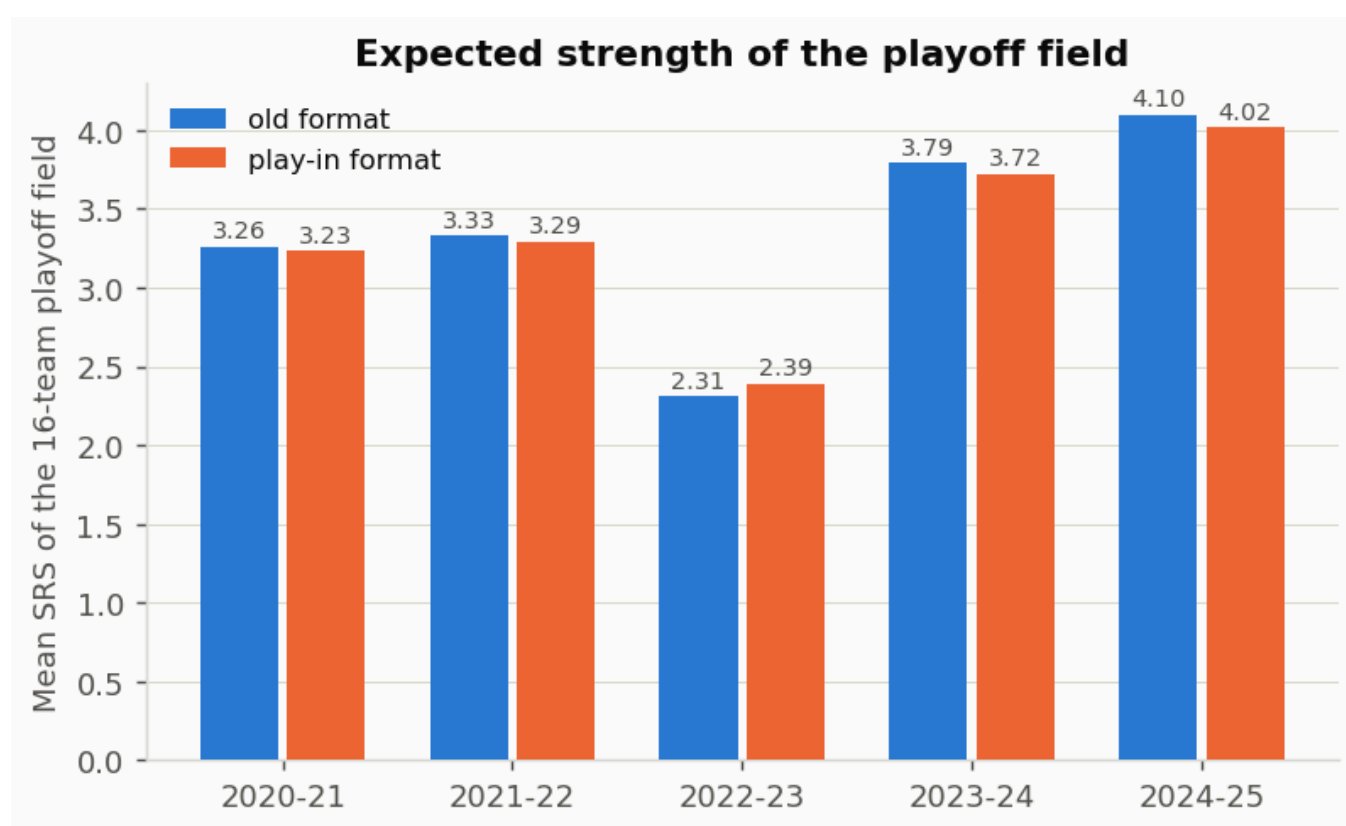
W is the number of simulated tournaments the highest-rated team won out of M. The interval is 95% around the simulated paired difference and reflects simulation noise only.

Expected champion and expected field

Season	E[champion SRS], old	E[champion SRS], play-in	Field SRS sum, old	Field SRS sum, play-in	Field mean, old	Field mean, play-in
2020-21	6.574	6.562	52.16	51.72	3.260	3.233
2021-22	5.695	5.674	53.29	52.65	3.331	3.291

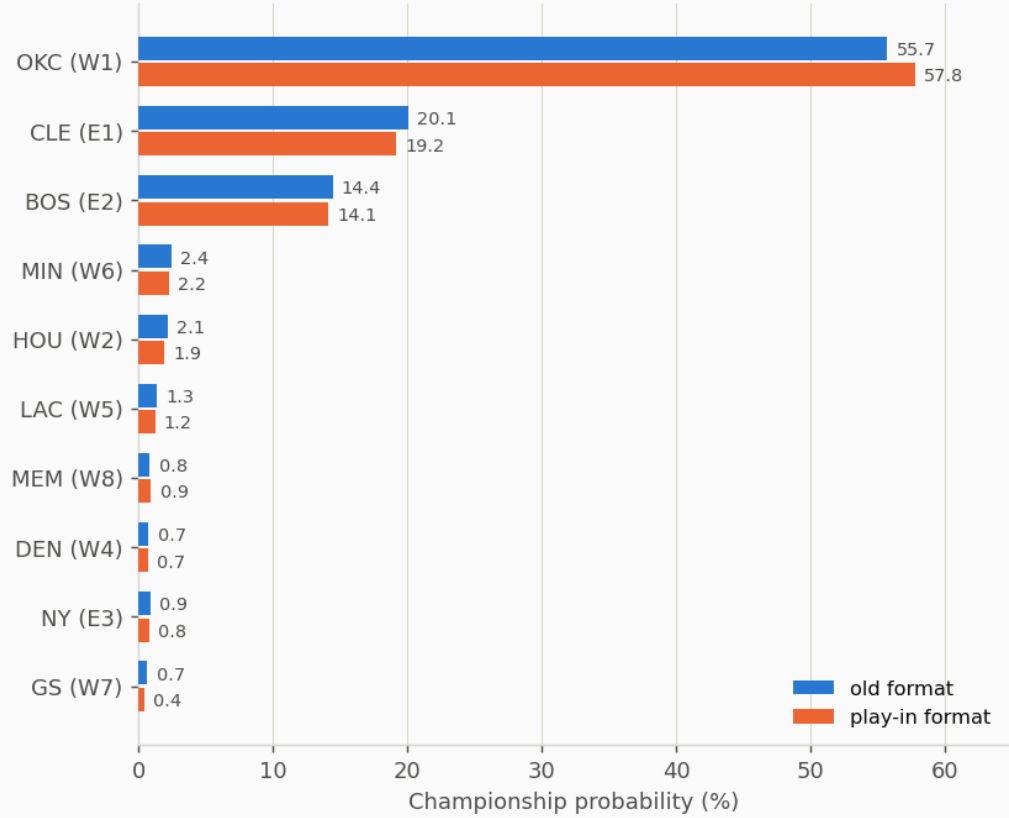
Season	E[champion SRS], old	E[champion SRS], play-in	Field SRS sum, old	Field SRS sum, play-in	Field mean, old	Field mean, play-in
2022-23	4.191	4.178	36.99	38.29	2.312	2.393
2023-24	8.602	8.611	60.65	59.58	3.790	3.724
2024-25	10.467	10.587	65.52	64.30	4.095	4.019

The playoff field's average rating falls slightly under the play-in format in four of five seasons and rises in 2022-23, when the teams in positions 9 and 10 were genuinely better than the ones in 7 and 8. Across all ten conference-seasons, a team in position 9 or 10 was rated above one of the 7/8 teams six times - which is why the format's effect has no fixed sign.

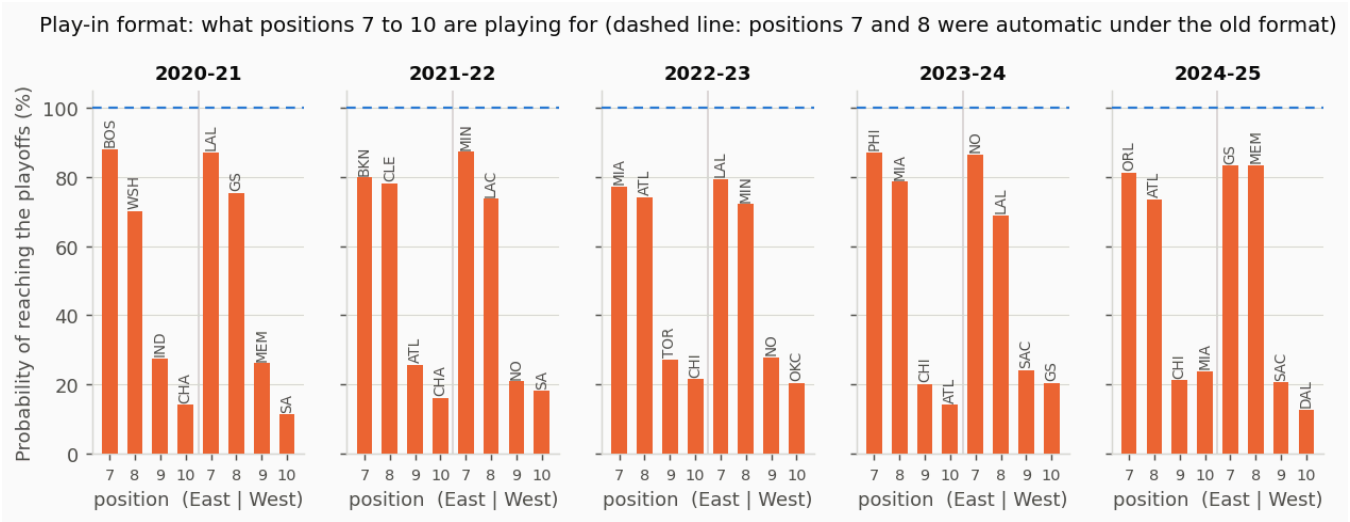


One season in detail: 2024-25

2024-25: ten highest-rated teams (conference and regular-season position in brackets)



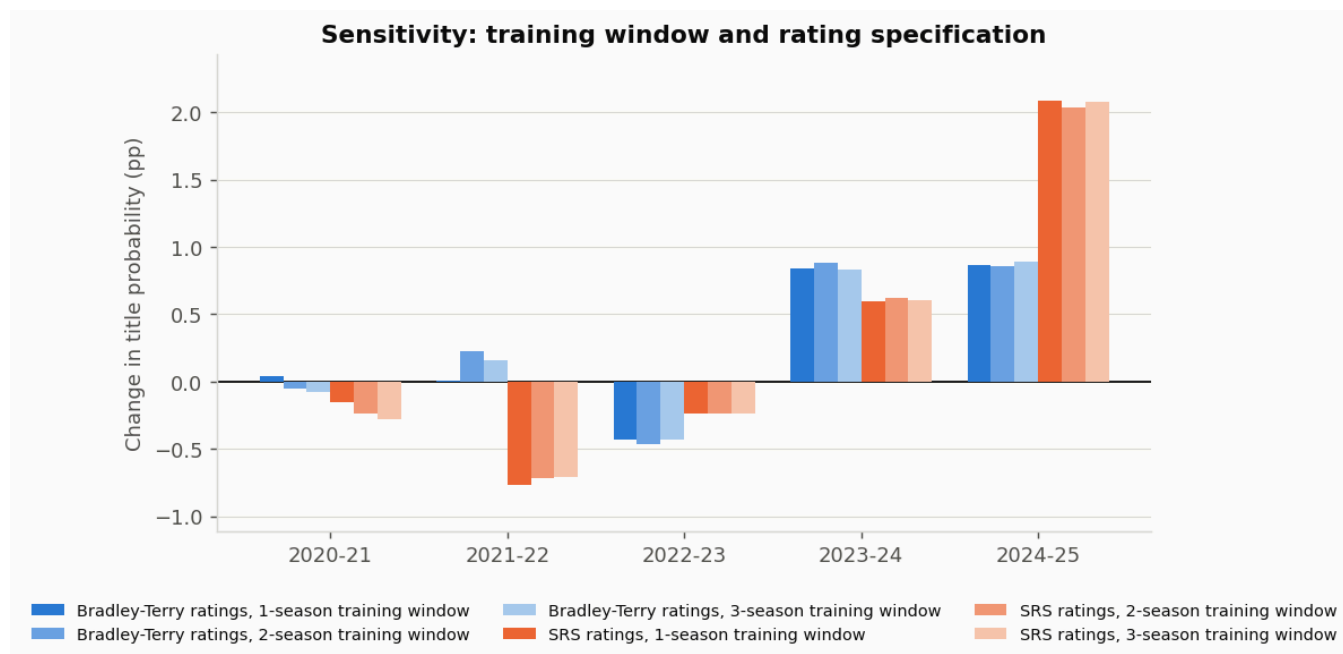
What the play-in teams are playing for



Positions 7 and 8 are no longer safe: across the five seasons, the probability that both keep their places averages 0.59 per conference. A position-9 team's chance of reaching the playoffs runs from 20% to 28%, and position 10's from 11% to 24%.

11. Sensitivity

Training window and rating specification



Season	Bradley-Terry ratings, 1-season training window	Bradley-Terry ratings, 2-season training window	Bradley-Terry ratings, 3-season training window	SRS ratings, 1-season training window	SRS ratings, 2-season training window	SRS ratings, 3-season training window
2020-21	+0.04	-0.05	-0.08	-0.15	-0.24	-0.28
2021-22	+0.01	+0.23	+0.16	-0.77	-0.71	-0.71
2022-23	-0.43	-0.46	-0.43	-0.24	-0.23	-0.24
2023-24	+0.84	+0.88	+0.83	+0.59	+0.62	+0.61
2024-25	+0.87	+0.86	+0.89	+2.09	+2.03	+2.08

Across every window and both rating definitions, the effect stays inside $[-0.77, +2.09]$ percentage points. Within a rating definition, changing the training window almost never changes a season's sign: the only exception is Bradley-Terry ratings in 2020-21, where every estimate is within 0.08 pp of zero. The window is not driving the answer.

When the alternative rating disagrees about who is strongest

The Bradley-Terry specification fits ratings to wins and losses only, ignoring margins, with a ridge penalty so early-season fits stay finite. It names a different strongest team in two seasons. Both readings are reported: the effect on the team **this** model calls strongest, and the effect on the team the baseline SRS model calls strongest.

Season	Target team	Which target	Old	Play-in	Difference (pp)
2020-21	UTAH	this model's strongest team	0.2527	0.2520	-0.08
2020-21	UTAH	baseline SRS strongest team	0.2527	0.2520	-0.08

Season	Target team	Which target	Old	Play-in	Difference (pp)
2021-22	PHX	this model's strongest team	0.4692	0.4708	+0.16
2021-22	BOS	baseline SRS strongest team	0.0567	0.0563	-0.03
2022-23	MIL	this model's strongest team	0.2656	0.2613	-0.43
2022-23	BOS	baseline SRS strongest team	0.1964	0.2020	+0.56
2023-24	BOS	this model's strongest team	0.4941	0.5024	+0.83
2023-24	BOS	baseline SRS strongest team	0.4941	0.5024	+0.83
2024-25	OKC	this model's strongest team	0.5350	0.5439	+0.89
2024-25	OKC	baseline SRS strongest team	0.5350	0.5439	+0.89

This is the single most important sensitivity in the study. In 2022-23 the two ratings disagree both about who is strongest and about the sign of the effect. "Strongest" is itself an estimate, and the answer moves when that estimate moves.

Finals home court

Four rules were tried: the published NBA chain, the workbook's 50/50 pilot assumption, and forcing every tie to the Eastern or the Western team. The highest-rated team's title probability is identical to machine precision under all four, because that team is never in a tied pair. The expected champion rating moves by at most 0.0024 SRS points. The tie assumption is immaterial to this study's conclusion, and it no longer needs to be an assumption at all.

Data issues

Variant	Season	Team	Difference (pp)
ratings forced to the ESPN standings scoring totals	2020-21	UTAH	-0.275
ratings forced to the ESPN standings scoring totals	2021-22	BOS	-0.690
ratings forced to the ESPN standings scoring totals	2022-23	BOS	-0.237
ratings forced to the ESPN standings scoring totals	2023-24	BOS	+0.613
ratings forced to the ESPN standings scoring totals	2024-25	OKC	+2.079
archive date defect corrected (game 401161536 moved to 2020-02-29)	2020-21	UTAH	-0.276
archive date defect corrected (game 401161536 moved to 2020-02-29)	2021-22	BOS	-0.710
archive date defect corrected (game 401161536 moved to 2020-02-29)	2022-23	BOS	-0.237
archive date defect corrected (game 401161536 moved to 2020-02-29)	2023-24	BOS	+0.605
archive date defect corrected (game 401161536 moved to 2020-02-29)	2024-25	OKC	+2.079

Neither of the audit's two open data items - the disputed standings totals and the one mis-dated game - moves any season's result by as much as 0.020 percentage points.

12. Limitations

1. **Standings and ratings are frozen, so this is not the policy's effect.** A real play-in changes what teams do in March: who rests, who tanks, who fights for seventh instead of settling for ninth. None of that is in the model. The number here is the bracket's mechanical effect, and it is a lower bound on the amount of uncertainty about the policy, not a measure of it.
 2. **"Strongest" is an estimate, not a fact.** The baseline calls Boston the strongest team three times; a win-loss rating disagrees twice and flips the sign of the effect in 2022-23. Any conclusion about "the best team" inherits the rating's error.
 3. **The game model is deliberately simple.** One rating difference, one home term, games independent given ratings. No injuries, no rest, no travel, no matchup effects, no momentum. It is well calibrated on held-out games, which is the most that can be claimed for it.
 4. **Five seasons is five observations.** The play-in has existed in this form only since 2020-21. The five-season average is an average over the five seasons that exist, not an estimate with a sampling distribution, and the Monte Carlo interval around it does not represent that uncertainty.
-

13. Conclusion

Over the five seasons in which the play-in tournament has existed, and holding standings and team strengths fixed, the format changed the highest-rated team's probability of winning the championship by **+0.29 percentage points on average** - positive in 2 seasons, negative in 3, and larger than one percentage point only in 2024-25.

That is a small effect, and its sign is not a property of the format. It depends on whether the play-in happens to promote a stronger or a weaker team into the seed the best team must beat, which changes season by season. The honest summary is that **the play-in tournament does not systematically help or hurt the best team**; it adds a small amount of noise whose direction depends on the season's standings.

Nothing here supports a claim that the play-in is better or worse overall. It is a comparison on one metric, inside one model, with the rest of the policy held fixed.

14. Plain-English guide to every computation

Scoring margin. Points scored minus points allowed in a game. Average them over a season and you have a crude strength measure.

SRS. The crude measure is unfair to teams with hard schedules. SRS fixes that by defining a team's rating as its average margin plus the average rating of its opponents. Every team's rating depends on every other team's, so all 30 are solved together as a system of equations, anchored by making them sum to zero. A rating of +5 means "about five points better than an average team against an average schedule".

Pregame ratings. To test whether the model predicts, you must not let it see the answer. So for each game date the ratings are rebuilt from games played on earlier dates only.

The logistic model. Take the rating gap, multiply by `beta` to turn points into log-odds, add `h` if the team is home (subtract it if away), then squash into a probability with the logistic function. Fitted by maximum likelihood on past seasons: choose the `beta` and `h` that make what actually happened as likely as possible.

Backward recursion for a series. Work backwards from the end. If you have four wins you have won (probability 1); four losses and you have lost (probability 0). Otherwise your chance from a given score is your chance of winning the next game times your chance from one-win-better, plus your chance of losing it times your chance from one-loss-worse. Because home court changes game by game, this is exact where a binomial formula would not be.

Why the six play-in outcomes cannot be multiplied. The team that wins the 7/8 game becomes seed 7 and is therefore not available to be seed 8. Seed 7 and seed 8 are dependent. Multiplying "chance of being seed 7" by "chance of being seed 8" would count impossible combinations.

Propagating through the bracket. After the first round you do not know who won, so you carry a probability distribution over possible winners. Each following round combines two such distributions: for every possible pairing, the chance they meet times the chance each wins.

Qualification probabilities summing to two. Two of the four play-in teams reach the playoffs, always. So their four probabilities must add to exactly 2 - a check that catches most implementation mistakes instantly.

Monte Carlo. Instead of computing, play the season out with random numbers hundreds of thousands of times and count. If simulation and exact calculation agree, two independent implementations of the rules agree, which is much stronger evidence than either alone.

Common random numbers. Both formats are run on the same random draws, so a difference between them is a real difference in the rules rather than a difference in luck. It shrinks the noise in the comparison by a factor of about 6.

Brier score and log loss. Two ways of scoring probability forecasts. Brier is the average squared error between the forecast and what happened; log loss punishes confident mistakes harder. Lower is better for both.

Calibration. Of the games you called 70%, did about 70% happen? Accuracy alone is not enough here, because these probabilities get multiplied together through four rounds.

15. Corrections for Pages 1 to 5

Every number on the handwritten pages was recomputed from the raw games

(`outputs/tables/worked_examples.csv`). **All of them are arithmetically correct - no figure needs changing.** Four labelling changes are needed so the pages say what they mean:

1. **Page 2 - label the ratings.** The four solved values (BOS 6.106, CLE 3.375, NY -6.894, IND -2.587) come from a 22-game mini-league among those four teams only. Write "restricted four-team ratings, not full-season NBA SRS" on the page. Boston's full-season 2024-25 SRS is 8.28, not 6.11.
2. **Page 3 - state the H convention.** Write the equation as $P(A \text{ beats } B) = \text{sigmoid}(\beta \cdot (R_A - R_B) + h \cdot H)$ with $H = +1, -1, 0$, and add one line: this is the same model as the fitted form "P(home wins) with $H = 1$ at home, 0 at neutral", seen from team A instead of from the host. Without that line the two forms look like different models.
3. **Pages 4 and 5 - say what 83.22% is.** It is the probability Golden State **qualifies** for the playoffs, as seed 7 or seed 8. It is not the probability of winning a play-in game (that is 0.5207 against Memphis at home). Add: "seed 7 and seed 8 are dependent - use the six ordered outcomes, never the product of two marginals."

4. **Page 6 - the target is 2, not 1.** Label the check "the four qualification probabilities sum to 2, because exactly two of the four teams get in". A reader who expects probabilities to sum to 1 will think the page is wrong.

One presentational note for Page 5: Memphis's qualification probability (0.8338) is slightly **higher** than Golden State's (0.8322) even though Golden State hosts, because Memphis is the better-rated team. That is worth a sentence on the page; it looks like an error and is not.

Worked-example verification

page	quantity	value_on_the_page	recomputed	agrees
1	average scoring margin	11.400000	11.400000	True
2	games inside the four-team subset	22.000000	22	True
2	BOS restricted rating	6.105769	6.105769	True
2	CLE restricted rating	3.375000	3.375000	True
2	NY restricted rating	-6.894231	-6.894231	True
2	IND restricted rating	-2.586538	-2.586538	True
2	ratings sum to zero	0.000000	-0.000000	True
3	beta (3-season window ending 2023-24)	0.112891	0.112891	True
3	h	0.222492	0.222492	True
3	P(Golden State beats Memphis at home)	0.520743	0.520743	True
4	P(seed 7 = GS, seed 8 = MEM)	0.354538	0.354538	True
4	P(seed 7 = GS, seed 8 = SAC)	0.102452	0.102452	True
4	P(seed 7 = GS, seed 8 = DAL)	0.063753	0.063753	True
4	P(seed 7 = MEM, seed 8 = GS)	0.311435	0.311435	True
4	P(seed 7 = MEM, seed 8 = SAC)	0.103265	0.103265	True
4	P(seed 7 = MEM, seed 8 = DAL)	0.064557	0.064557	True
4	six ordered outcomes sum to 1	1.000000	1.000000	True
5	P(GS qualifies)	0.832178	0.832178	True
5	P(MEM qualifies)	0.833795	0.833795	True
5	P(SAC qualifies)	0.205717	0.205717	True
5	P(DAL qualifies)	0.128310	0.128310	True
6	the four qualification probabilities sum to 2	2.000000	2.000000	True
8	P(Golden State beats Memphis, neutral court)	0.465188	0.465188	True
8	constant-p best-of-seven	0.424217	0.424217	True
8	backward recursion agrees with the binomial sum	0.424217	0.424217	True

Appendix A - what runs what

File	What it does
<code>run_all.py</code>	one command, whole study
<code>src/nbaplayin/audit.py</code>	step 1, workbook against independent sources
<code>src/nbaplayin/ratings.py</code>	step 2a, SRS and Bradley-Terry, full-season and pregame
<code>src/nbaplayin/model.py</code>	step 2b, the logistic game model
<code>src/nbaplayin/tournament.py</code>	steps 3-4, both formats, exact
<code>src/nbaplayin/montecarlo.py</code>	step 5, simulation
<code>src/nbaplayin/validation.py</code>	step 6a, held-out prediction
<code>src/nbaplayin/sensitivity.py</code>	step 6b, every variant
<code>src/nbaplayin/worked_examples.py</code>	recomputes the six handwritten pages
<code>src/nbaplayin/figures.py</code>	the seven charts
<code>src/nbaplayin/report.py</code>	this document
<code>tests/</code>	5 test files, run with <code>pytest</code>

Appendix B - fitted coefficients, every window and specification

season	spec	window	train_first	train_last	train_games	beta	beta_se	h	h_se	role
2021	srs	1	2020	2020	750	0.09791	0.01153	0.18753	0.08318	sensitivity window
2021	srs	2	2019	2020	1672	0.10818	0.00823	0.31786	0.05446	sensitivity window
2021	srs	3	2018	2020	2593	0.11482	0.00699	0.33780	0.04338	prespecified baseline
2022	srs	1	2021	2021	769	0.11535	0.01297	0.23867	0.07733	sensitivity window
2022	srs	2	2020	2021	1519	0.10585	0.00863	0.21514	0.05659	sensitivity window
2022	srs	3	2019	2021	2441	0.11016	0.00695	0.29179	0.04451	prespecified baseline
2023	srs	1	2022	2022	922	0.11113	0.01189	0.17674	0.06995	sensitivity window
2023	srs	2	2021	2022	1691	0.11304	0.00876	0.20467	0.05187	sensitivity window
2023	srs	3	2020	2022	2441	0.10765	0.00698	0.20000	0.04399	prespecified baseline
2024	srs	1	2023	2023	922	0.10707	0.01410	0.31284	0.06923	sensitivity window
2024	srs	2	2022	2023	1844	0.10941	0.00908	0.24598	0.04917	sensitivity window
2024	srs	3	2021	2023	2613	0.11139	0.00744	0.24392	0.04149	prespecified baseline

season	spec	window	train_first	train_last	train_games	beta	beta_se	h	h_se	role
2025	srs	1	2024	2024	923	0.11727	0.01053	0.17186	0.07211	sensitivity window
2025	srs	2	2023	2024	1845	0.11386	0.00842	0.24574	0.04992	sensitivity window
2025	srs	3	2022	2024	2767	0.11289	0.00687	0.22249	0.04063	prespecified baseline
2021	bt	1	2020	2020	750	0.80267	0.09318	0.18461	0.08325	sensitivity window
2021	bt	2	2019	2020	1672	0.90644	0.06880	0.31456	0.05441	sensitivity window
2021	bt	3	2018	2020	2593	0.94492	0.05726	0.33422	0.04335	prespecified baseline
2022	bt	1	2021	2021	769	1.09777	0.12613	0.23221	0.07704	sensitivity window
2022	bt	2	2020	2021	1519	0.91300	0.07539	0.21039	0.05642	sensitivity window
2022	bt	3	2019	2021	2441	0.95041	0.06050	0.28736	0.04439	prespecified baseline
2023	bt	1	2022	2022	922	0.92370	0.10197	0.17696	0.06968	sensitivity window
2023	bt	2	2021	2022	1691	0.99466	0.07940	0.20204	0.05166	sensitivity window
2023	bt	3	2020	2022	2441	0.91668	0.06062	0.19715	0.04385	prespecified baseline
2024	bt	1	2023	2023	922	0.91210	0.11728	0.30868	0.06929	sensitivity window
2024	bt	2	2022	2023	1844	0.91868	0.07694	0.24353	0.04911	sensitivity window
2024	bt	3	2021	2023	2613	0.96876	0.06569	0.24041	0.04140	prespecified baseline
2025	bt	1	2024	2024	923	0.99916	0.09026	0.16266	0.07205	sensitivity window
2025	bt	2	2023	2024	1845	0.96839	0.07128	0.23898	0.04992	sensitivity window
2025	bt	3	2022	2024	2767	0.95358	0.05838	0.21799	0.04058	prespecified baseline

Appendix C - validation, every season and window

season	window	test_set	games	brier	log_loss	accuracy	brier_baseline	log_loss_baseline
2021	1	Regular season (pregame ratings)	769	0.2191	0.6285	0.6502	0.2475	0.6881

season	window	test_set	games	brier	log_loss	accuracy	brier_baseline	log_loss_baseline
2021	1	Play-in games (frozen ratings)	6	0.1964	0.5844	0.8333	0.2266	0.6463
2021	1	Playoffs (frozen ratings)	85	0.2282	0.6485	0.6588	0.2466	0.6863
2021	2	Regular season (pregame ratings)	769	0.2191	0.6280	0.6359	0.2474	0.6879
2021	2	Play-in games (frozen ratings)	6	0.1796	0.5486	0.8333	0.2104	0.6135
2021	2	Playoffs (frozen ratings)	85	0.2274	0.6468	0.6471	0.2458	0.6848
2021	3	Regular season (pregame ratings)	769	0.2192	0.6282	0.6359	0.2476	0.6883
2021	3	Play-in games (frozen ratings)	6	0.1764	0.5415	0.8333	0.2073	0.6073
2021	3	Playoffs (frozen ratings)	85	0.2271	0.6460	0.6471	0.2459	0.6849
2022	1	Regular season (pregame ratings)	922	0.2218	0.6354	0.6529	0.2487	0.6905
2022	1	Play-in games (frozen ratings)	6	0.2370	0.6667	0.6667	0.2352	0.6635
2022	1	Playoffs (frozen ratings)	87	0.2263	0.6447	0.6552	0.2425	0.6781
2022	2	Regular season (pregame ratings)	922	0.2219	0.6352	0.6508	0.2485	0.6902
2022	2	Play-in games (frozen ratings)	6	0.2378	0.6682	0.6667	0.2369	0.6669
2022	2	Playoffs (frozen ratings)	87	0.2272	0.6467	0.6552	0.2432	0.6795
2022	3	Regular season (pregame ratings)	922	0.2224	0.6364	0.6497	0.2490	0.6911
2022	3	Play-in games (frozen ratings)	6	0.2345	0.6616	0.6667	0.2333	0.6595
2022	3	Playoffs (frozen ratings)	87	0.2258	0.6437	0.6667	0.2418	0.6766
2023	1	Regular season (pregame ratings)	922	0.2294	0.6506	0.6226	0.2461	0.6853
2023	1	Play-in games (frozen ratings)	6	0.2633	0.7199	0.5000	0.2515	0.6962
2023	1	Playoffs (frozen ratings)	84	0.2445	0.6829	0.6071	0.2441	0.6813
2023	2	Regular season (pregame ratings)	922	0.2291	0.6498	0.6269	0.2457	0.6846
2023	2	Play-in games (frozen ratings)	6	0.2641	0.7214	0.5000	0.2520	0.6973

season	window	test_set	games	brier	log_loss	accuracy	brier_baseline	log_loss_baseline
2023	2	Playoffs (frozen ratings)	84	0.2440	0.6821	0.5833	0.2434	0.6800
2023	3	Regular season (pregame ratings)	922	0.2291	0.6498	0.6291	0.2458	0.6848
2023	3	Play-in games (frozen ratings)	6	0.2634	0.7201	0.5000	0.2518	0.6969
2023	3	Playoffs (frozen ratings)	84	0.2437	0.6813	0.5833	0.2437	0.6804
2024	1	Regular season (pregame ratings)	923	0.2117	0.6099	0.6479	0.2503	0.6939
2024	1	Play-in games (frozen ratings)	6	0.2000	0.5925	0.8333	0.2082	0.6091
2024	1	Playoffs (frozen ratings)	82	0.2190	0.6293	0.6707	0.2429	0.6790
2024	2	Regular season (pregame ratings)	923	0.2109	0.6080	0.6511	0.2494	0.6920
2024	2	Play-in games (frozen ratings)	6	0.2072	0.6072	0.8333	0.2166	0.6261
2024	2	Playoffs (frozen ratings)	82	0.2192	0.6295	0.6951	0.2437	0.6804
2024	3	Regular season (pregame ratings)	923	0.2108	0.6079	0.6522	0.2494	0.6920
2024	3	Play-in games (frozen ratings)	6	0.2073	0.6076	0.8333	0.2169	0.6268
2024	3	Playoffs (frozen ratings)	82	0.2191	0.6292	0.6951	0.2437	0.6805
2025	1	Regular season (pregame ratings)	919	0.2127	0.6117	0.6485	0.2490	0.6911
2025	1	Play-in games (frozen ratings)	6	0.2249	0.6407	0.8333	0.2511	0.6953
2025	1	Playoffs (frozen ratings)	84	0.2370	0.6665	0.6429	0.2464	0.6860
2025	2	Regular season (pregame ratings)	919	0.2130	0.6125	0.6507	0.2493	0.6918
2025	2	Play-in games (frozen ratings)	6	0.2281	0.6467	0.5000	0.2526	0.6984
2025	2	Playoffs (frozen ratings)	84	0.2363	0.6646	0.5952	0.2453	0.6837
2025	3	Regular season (pregame ratings)	919	0.2128	0.6121	0.6485	0.2492	0.6915
2025	3	Play-in games (frozen ratings)	6	0.2274	0.6454	0.5000	0.2522	0.6976
2025	3	Playoffs (frozen ratings)	84	0.2362	0.6644	0.5952	0.2455	0.6841

Appendix D - play-in seed states, all five seasons

season	conference	state	seed7	seed8	probability
2021	Eastern	1	BOS	WSH	0.370130
2021	Eastern	2	BOS	IND	0.196775
2021	Eastern	3	BOS	CHA	0.101423
2021	Eastern	4	WSH	BOS	0.212476
2021	Eastern	5	WSH	IND	0.079135
2021	Eastern	6	WSH	CHA	0.040061
2021	Western	1	LAL	GS	0.383288
2021	Western	2	LAL	MEM	0.171385
2021	Western	3	LAL	SA	0.074617
2021	Western	4	GS	LAL	0.242213
2021	Western	5	GS	MEM	0.089883
2021	Western	6	GS	SA	0.038614
2022	Eastern	1	BKN	CLE	0.321577
2022	Eastern	2	BKN	ATL	0.134102
2022	Eastern	3	BKN	CHA	0.083606
2022	Eastern	4	CLE	BKN	0.259667
2022	Eastern	5	CLE	ATL	0.123665
2022	Eastern	6	CLE	CHA	0.077384
2022	Western	1	MIN	LAC	0.373578
2022	Western	2	MIN	NO	0.140838
2022	Western	3	MIN	SA	0.122246
2022	Western	4	LAC	MIN	0.236218
2022	Western	5	LAC	NO	0.067862
2022	Western	6	LAC	SA	0.059259
2023	Eastern	1	MIA	ATL	0.278790
2023	Eastern	2	MIA	TOR	0.144766
2023	Eastern	3	MIA	CHI	0.114247
2023	Eastern	4	ATL	MIA	0.233997
2023	Eastern	5	ATL	TOR	0.127528
2023	Eastern	6	ATL	CHI	0.100673
2023	Western	1	LAL	MIN	0.288045
2023	Western	2	LAL	NO	0.160797
2023	Western	3	LAL	OKC	0.118119
2023	Western	4	MIN	LAL	0.227515
2023	Western	5	MIN	NO	0.118549

season	conference	state	seed7	seed8	probability
2023	Western	6	MIN	OKC	0.086976
2024	Eastern	1	PHI	MIA	0.385524
2024	Eastern	2	PHI	CHI	0.125518
2024	Eastern	3	PHI	ATL	0.087921
2024	Eastern	4	MIA	PHI	0.272239
2024	Eastern	5	MIA	CHI	0.075818
2024	Eastern	6	MIA	ATL	0.052980
2024	Western	1	NO	LAL	0.338924
2024	Western	2	NO	SAC	0.168512
2024	Western	3	NO	GS	0.143056
2024	Western	4	LAL	NO	0.214367
2024	Western	5	LAL	SAC	0.072928
2024	Western	6	LAL	GS	0.062213
2025	Eastern	1	ORL	ATL	0.310529
2025	Eastern	2	ORL	CHI	0.124808
2025	Eastern	3	ORL	MIA	0.139869
2025	Eastern	4	ATL	ORL	0.237780
2025	Eastern	5	ATL	CHI	0.087983
2025	Eastern	6	ATL	MIA	0.099031
2025	Western	1	GS	MEM	0.354538
2025	Western	2	GS	SAC	0.102452
2025	Western	3	GS	DAL	0.063753
2025	Western	4	MEM	GS	0.311435
2025	Western	5	MEM	SAC	0.103265
2025	Western	6	MEM	DAL	0.064557

Appendix E - Finals hosting ties and how each was resolved

season	east	west	rule	east_hosts
2021	CHA	SA	record against the opposite conference	0.0
2022	MIA	GS	head-to-head record	0.0
2022	TOR	DEN	record against the opposite conference	0.0
2022	CHI	MIN	head-to-head record	1.0
2023	CLE	MEM	record against the opposite conference	0.0
2023	BKN	PHX	head-to-head record	0.0
2023	MIA	LAC	head-to-head record	1.0
2023	MIA	GS	record against the opposite conference	1.0

season	east	west	rule	east_hosts
2023	CHI	OKC	head-to-head record	0.0
2024	NY	DAL	head-to-head record	0.0
2024	MIL	PHX	record against the opposite conference	0.0
2024	MIL	NO	record against the opposite conference	0.0
2024	ORL	LAL	record against the opposite conference	0.0
2024	IND	LAL	record against the opposite conference	0.0
2024	PHI	LAL	record against the opposite conference	0.0
2024	MIA	SAC	head-to-head record	1.0
2024	MIA	GS	record against the opposite conference	0.0
2025	IND	LAL	head-to-head record	0.0
2025	IND	DEN	record against the opposite conference	1.0
2025	IND	LAC	head-to-head record	1.0
2025	MIL	GS	head-to-head record	0.0
2025	MIL	MEM	head-to-head record	0.0
2025	ATL	SAC	record against the opposite conference	0.0
2025	CHI	DAL	head-to-head record	0.0

Appendix F - per-team results

The full 150-row table of ratings, qualification probabilities and title probabilities under both formats is in `outputs/tables/exact_team_results.csv` , and round-by-round advancement probabilities are in `outputs/tables/advancement.csv` .